

continuous on $\overline{\Omega}$. Moreover $|F(z)| = 1$ on the two boundary lines $z = x$ and $z = iy$. However, $F(z)$ is unbounded in Ω , since for example, we have $F(z) = e^{r^2}$ if $z = r\sqrt{i} = re^{i\pi/4}$.

5 Homotopies and simply connected domains

The key to the general form of Cauchy's theorem, as well as the analysis of multiple-valued functions, is to understand in what regions we can define the primitive of a given holomorphic function. Note the relevance to the study of the logarithm, which arises as a primitive of $1/z$. The question is not just a local one, but is also global in nature. Its elucidation requires the notion of homotopy, and the resulting idea of simple-connectivity.

Let γ_0 and γ_1 be two curves in an open set Ω with common end-points. So if $\gamma_0(t)$ and $\gamma_1(t)$ are two parametrizations defined on $[a, b]$, we have

$$\gamma_0(a) = \gamma_1(a) = \alpha \quad \text{and} \quad \gamma_0(b) = \gamma_1(b) = \beta.$$

These two curves are said to be **homotopic in Ω** if for each $0 \leq s \leq 1$ there exists a curve $\gamma_s \subset \Omega$, parametrized by $\gamma_s(t)$ defined on $[a, b]$, such that for every s

$$\gamma_s(a) = \alpha \quad \text{and} \quad \gamma_s(b) = \beta,$$

and for all $t \in [a, b]$

$$\gamma_s(t)|_{s=0} = \gamma_0(t) \quad \text{and} \quad \gamma_s(t)|_{s=1} = \gamma_1(t).$$

Moreover, $\gamma_s(t)$ should be jointly continuous in $s \in [0, 1]$ and $t \in [a, b]$.

Loosely speaking, two curves are homotopic if one curve can be deformed into the other by a continuous transformation without ever leaving Ω (Figure 6).

Theorem 5.1 *If f is holomorphic in Ω , then*

$$\int_{\gamma_0} f(z) dz = \int_{\gamma_1} f(z) dz$$

whenever the two curves γ_0 and γ_1 are homotopic in Ω .

Proof. The key to the proof lies in showing that if two curves are close to each other and have the same end-points, then the integrals over these curves are equal. Recall that by definition, the function $F(s, t) = \gamma_s(t)$ is continuous on $[0, 1] \times [a, b]$. In particular, since the image of F , which we

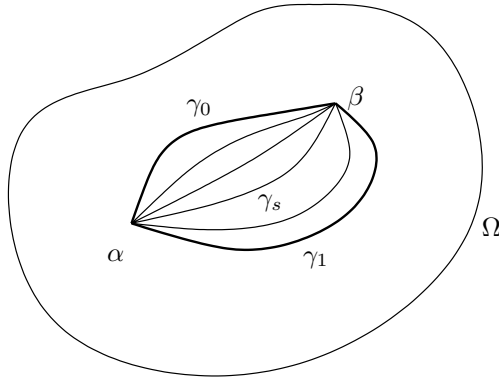


Figure 6. Homotopy of curves

denote by K , is compact, there exists $\epsilon > 0$ such that every disc of radius 3ϵ centered at a point in the image of F is completely contained in Ω . If not, for every $\ell \geq 0$, there exist points $z_\ell \in K$ and w_ℓ in the complement of Ω such that $|z_\ell - w_\ell| < 1/\ell$. By compactness of K , there exists a subsequence of $\{z_\ell\}$, say $\{z_{\ell_k}\}$, that converges to a point $z \in K \subset \Omega$. By construction, we must have $w_{\ell_k} \rightarrow z$ as well, and since $\{w_\ell\}$ lies in the complement of Ω which is closed, we must have $z \in \Omega^c$ as well. This is a contradiction.

Having found an ϵ with the desired property, we may, by the uniform continuity of F , select δ so that

$$\sup_{t \in [a, b]} |\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon \quad \text{whenever } |s_1 - s_2| < \delta.$$

Fix s_1 and s_2 with $|s_1 - s_2| < \delta$. We then choose discs $\{D_0, \dots, D_n\}$ of radius 2ϵ , and consecutive points $\{z_0, \dots, z_{n+1}\}$ on γ_{s_1} and $\{w_0, \dots, w_{n+1}\}$ on γ_{s_2} such that the union of these discs covers both curves, and

$$z_i, z_{i+1}, w_i, w_{i+1} \in D_i.$$

The situation is illustrated in Figure 7.

Also, we choose $z_0 = w_0$ as the beginning end-point of the curves and $z_{n+1} = w_{n+1}$ as the common end-point. On each disc D_i , let F_i denote a primitive of f (Theorem 2.1, Chapter 2). On the intersection of D_i and D_{i+1} , F_i and F_{i+1} are two primitives of the same function, so they must differ by a constant, say c_i . Therefore

$$F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1}),$$

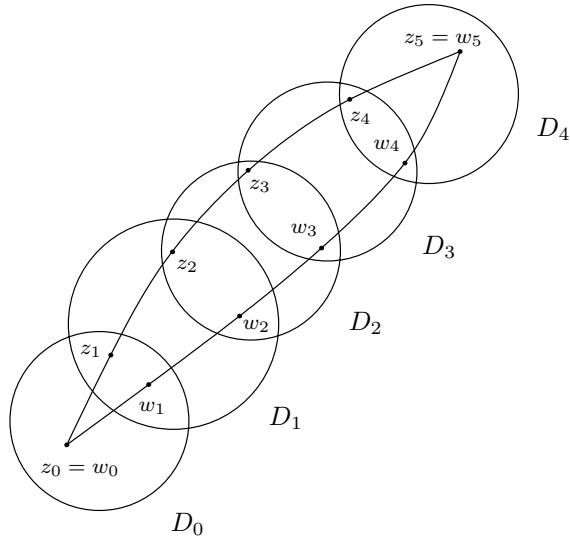


Figure 7. Covering two nearby curves with discs

hence

$$(5) \quad F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}).$$

This implies

$$\begin{aligned} \int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f &= \sum_{i=0}^n [F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^n [F_i(w_{i+1}) - F_i(w_i)] \\ &= \sum_{i=0}^n F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i)) \\ &= F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)), \end{aligned}$$

because of the cancellations due to (5). Since γ_{s_1} and γ_{s_2} have the same beginning and end point, we have proved that

$$\int_{\gamma_{s_1}} f = \int_{\gamma_{s_2}} f.$$

By subdividing $[0, 1]$ into subintervals $[s_i, s_{i+1}]$ of length less than δ , we may go from γ_0 to γ_1 by finitely many applications of the above argument, and the theorem is proved.

A region Ω in the complex plane is **simply connected** if any two pair of curves in Ω with the same end-points are homotopic.

EXAMPLE 1. A disc D is simply connected. In fact, if $\gamma_0(t)$ and $\gamma_1(t)$ are two curves lying in D , we can define $\gamma_s(t)$ by

$$\gamma_s(t) = (1 - s)\gamma_0(t) + s\gamma_1(t).$$

Note that if $0 \leq s \leq 1$, then for each t , the point $\gamma_s(t)$ is on the segment joining $\gamma_0(t)$ and $\gamma_1(t)$, and so is in D . The same argument works if D is replaced by a rectangle, or more generally by any open convex set. (See Exercise 21.)

EXAMPLE 2. The slit plane $\Omega = \mathbb{C} - \{(-\infty, 0]\}$ is simply connected. For a pair of curves γ_0 and γ_1 in Ω , we write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ ($j = 0, 1$) with $r_j(t)$ continuous and strictly positive, and $\theta_j(t)$ continuous with $|\theta_j(t)| < \pi$. Then, we can define $\gamma_s(t)$ as $r_s(t)e^{i\theta_s(t)}$ where

$$r_s(t) = (1 - s)r_0(t) + sr_1(t) \quad \text{and} \quad \theta_s(t) = (1 - s)\theta_0(t) + s\theta_1(t).$$

We then have $\gamma_s(t) \in \Omega$ whenever $0 \leq s \leq 1$.

EXAMPLE 3. With some effort one can show that the interior of a toy contour is simply connected. This requires that we divide the interior into several subregions. A general form of the argument is given in Exercise 4.

EXAMPLE 4. In contrast with the previous examples, the punctured plane $\mathbb{C} - \{0\}$ is not simply connected. Intuitively, consider two curves with the origin between them. It is impossible to continuously pass from one curve to the other without going over 0. A rigorous proof of this fact requires one further result, and will be given shortly.

Theorem 5.2 *Any holomorphic function in a simply connected domain has a primitive.*

Proof. Fix a point z_0 in Ω and define

$$F(z) = \int_{\gamma} f(w) dw$$

where the integral is taken over any curve in Ω joining z_0 to z . This definition is independent of the curve chosen, since Ω is simply connected,

and if $\tilde{\gamma}$ is another curve in Ω joining z_0 and z , we would have

$$\int_{\gamma} f(w) dw = \int_{\tilde{\gamma}} f(w) dw$$

by Theorem 5.1. Now we can write

$$F(z+h) - F(z) = \int_{\eta} f(w) dw$$

where η is the line segment joining z and $z+h$. Arguing as in the proof of Theorem 2.1 in Chapter 2, we find that

$$\lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h} = f(z).$$

As a result, we obtain the following version of Cauchy's theorem.

Corollary 5.3 *If f is holomorphic in the simply connected region Ω , then*

$$\int_{\gamma} f(z) dz = 0$$

for any closed curve γ in Ω .

This is immediate from the existence of a primitive.

The fact that the punctured plane is not simply connected now follows rigorously from the observation that the integral of $1/z$ over the unit circle is $2\pi i$, and not 0.

6 The complex logarithm

Suppose we wish to define the logarithm of a non-zero complex number. If $z = re^{i\theta}$, and we want the logarithm to be the inverse to the exponential, then it is natural to set

$$\log z = \log r + i\theta.$$

Here and below, we use the convention that $\log r$ denotes the *standard*¹ logarithm of the positive number r . The trouble with the above definition is that θ is unique only up to an integer multiple of 2π . However,

¹By the standard logarithm, we mean the natural logarithm of positive numbers that appears in elementary calculus.